# fyM1: Teaching an Open Model to Chat in Everyday Malagasy

Version 0.1 — some results pending (v0.2)

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October 2026

**Abstract**

Open language models that cover Malagasy tend to answer like an official letter: formal, stiff, and often half-translated from French or English. People in Madagascar do not text like that. We fine-tune an open model, **pending**, with QLoRA on a single free GPU so that it replies the way a friend would: short, friendly, mixing in French, and using everyday spelling. We describe how we chose the base model, how we built and anonymised the training data, how we trained and trimmed the model, and what we measured. On **pending** fresh everyday prompts, the share of replies that drift into full French or English falls from **pending** (base) to **pending** (fyM1). A blind human comparison is still collecting votes and is reported as pending.

## 1 Motivation: the formal-versus-everyday gap

Malagasy is the national language of Madagascar, but most Malagasy text online is formal: news, Wikipedia, religious and government writing. Models trained on that text learn that register. When a user writes a short, casual message, they get back a polished paragraph that no one would send to a friend.

Everyday Malagasy looks different. French words are mixed in freely, including French verbs used with Malagasy grammar. Spelling is phonetic and shortened. Replies are short, with humour and emoji. For this project, that mixed style is the target, not an error. A reply only counts as drifting if it is entirely French or English with no Malagasy in it.

Our goal is modest and measurable: a model that is clearly better than its base model at everyday Malagasy chat, without losing much general Malagasy ability, trained and served at zero cost.

## 2 Choosing the base model

We shortlisted recent open-weight instruction models between 7B and 12B parameters that fit QLoRA training on a 16 GB T4 GPU. Each candidate was loaded in 4-bit and scored on: reading comprehension (Belebele, `plt_Latn`; Bandarkar et al., 2024); translation in both directions with English (FLORES+ devtest, chrF++; Goyal et al., 2022, NLLB Team et al., 2022, Burchell et al., 2024, Popović, 2017); tokenizer efficiency (tokens per Malagasy word on Wikipedia sentences); and a short everyday-chat sniff test rated by a native speaker (1–5). License was a tie-breaker: we prefer permissive licenses when quality is close. Table 1 shows the numbers. We chose **pending** (license: **pending**).

| Model | License | Belebele | chrF++ mg→en | chrF++ en→mg | Tokens/word | Natural (1–5) |
|---|---|---|---|---|---|---|
| **pending** | | | | | | |

Table 1: Base-model candidates, all loaded in 4-bit. Higher is better except tokens per word.

# 3 Data

We describe the data but do not share it. It contains private conversations.

**Real chats.** The main source is a personal Messenger export from a native speaker. We parsed **pending** threads into **pending** short multi-turn conversations, splitting on long gaps and merging back-to-back messages from the same person. We kept conversations where most turns are Malagasy or Malagasy–French (language ID with GlotLID; Kargaran et al., 2023). A native speaker reviewed samples and approved **pending** conversations.

**Formal slice.** **pending** short single-turn examples built from Malagasy Wikipedia, so facts and standard spelling survive. We keep this slice small so it does not pull the style back to formal.

**Identity and refusals.** **pending** short exchanges, written by the native speaker, that answer "who are you?" consistently and politely refuse harmful requests.

**Synthetic chats.** **pending** synthetic conversations. We only planned to add them if there were too few real chats; any synthetic chats come from an open-weight model run by us, never a closed API, and a native speaker checks a sample.

In total the model trained on **pending** examples.

**Privacy process.** Raw exports never left the owner's laptop. A local script replaced names and nicknames (also inside message text), phone numbers, emails, links and account handles before anything was uploaded. Only the cleaned, approved data went to a private dataset repository, used by the training notebook and never made public. Test prompts were written fresh, so the test set contains no private text and is public.

# 4 Method

We train with QLoRA [Dettmers et al., 2023]: the base model is frozen in 4-bit NF4 and we learn LoRA adapters [Hu et al., 2022] on all linear layers, with rank **pending** and alpha **pending**. We use Unsloth [Han et al., 2023] on a free Google Colab GPU (**pending**). Settings: learning rate **pending** with a cosine schedule, sequence length **pending**, effective batch size **pending**, **pending** epochs (**pending** steps, **pending** of training). Loss is computed on assistant turns only, in the base model's own chat template, with a fixed system prompt in Malagasy. Free sessions disconnect, so checkpoints are pushed to a private repository during training and a new session resumes from the latest one. Final training loss was **pending**; best held-out loss was **pending**.

## 5 Making the model smaller: vocabulary trimming

Much of a multilingual model's vocabulary is for scripts and languages fyM1 never uses. Following Ushio et al. [2023], we keep only the tokens seen in Malagasy, French and English text, plus all special and byte-fallback tokens, and drop the matching embedding and output rows. The trim ships only if it passes three gates: held-out Malagasy text round-trips through the new tokenizer unchanged; greedy answers to the test prompts are unchanged for almost all prompts; and Belebele stays within one point. Result: gates **pending**. Vocabulary **pending** → **pending** tokens; parameters **pending** → **pending**; download **pending** → **pending**; speed on the demo CPU **pending** → **pending** tokens/s.

Removing whole layers [Gromov et al., 2024] would save more, but needs healing with extra training that free GPU time cannot pay for. We leave it for v0.2.

## 6 Results

We generated answers from the base model and fyM1 to **pending** fresh everyday prompts with the same system prompt and sampling settings (Table 2).

| Measure | Base | fyM1 |
|---|---|---|
| Drift to full French/English (↓) | **pending** | **pending** |
| Naturalness, LLM judge, 1–5 (↑) | **pending** | **pending** |
| Correctness, LLM judge, 1–5 (↑) | **pending** | **pending** |
| Held-out chat perplexity (↓) | **pending** | **pending** |
| Belebele accuracy (↑) | **pending** | **pending** |
| FLORES+ chrF++ mg→en (↑) | **pending** | **pending** |
| FLORES+ chrF++ en→mg (↑) | **pending** | **pending** |

Table 2: Base model versus fyM1. Belebele and FLORES+ are regression checks: they should not fall much.

**Automatic judge.** An LLM judge (**pending**) scored both answer sets blind, with labels hidden and order shuffled, on a published 1–5 rubric for naturalness and correctness [Zheng et al., 2023]. Judge scores are a cheap signal, not a substitute for native speakers.

**Leak check.** We test whether fyM1 repeats training text: **pending** of its answers share an 8-word sequence with the training chats. The release gate was at most 1%.

**Blind human comparison (pending, v0.2).** Native speakers rate pairs of answers in a public "arena": same prompt, two unlabelled answers in random order, pick the more natural one. So far: **pending** votes from **pending** raters; fyM1 win rate **pending** (95% Wilson interval **pending**–**pending**; Wilson, 1927).

## 7 Limitations

- The style comes mostly from one person's chats and their contacts. It reflects Antananarivo-style everyday speech, not other regions or dialects.

- The human comparison is still running. Until it is in, the main evidence is automatic.
- An LLM judge may not understand Malagasy slang well; its scores can be wrong in either direction.
- The model is small and runs on a free CPU. It can be slow, get facts wrong, and invent things.
- We train only once or twice on free hardware, so settings are not carefully tuned.

## 8 Ethics, consent and privacy

The data owner confirmed that everyone in the conversations consented to their messages being used. Raw data never left the owner's machine; only anonymised, approved data was used for training, and it is not released. We check that the model does not repeat training text before release (Section 6). The model was trained with a small set of polite refusals and gets a basic safety check before release, but it is not a safety-tested product. No closed-API model outputs are in the training data.

## 9 What comes next (v0.2)

- Blind human comparison results from the arena, with confidence intervals.
- The layer-pruning experiment.
- Longer training and more everyday data.

## References

Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabsa. The belebele benchmark: a parallel reading comprehension dataset in 122 language variants. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 749–775, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.44. URL `https://aclanthology.org/2024.acl-long.44/`.

Laurie Burchell, Jean Maillard, Antonios Anastasopoulos, Christian Federmann, Philipp Koehn, and Skyler Wang. Findings of the WMT 2024 shared task of the open language data initiative. In Barry Haddow, Tom Kocmi, Philipp Koehn, and Christof Monz, editors, *Proceedings of the Ninth Conference on Machine Translation*, pages 110–117, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.wmt-1.4. URL `https://aclanthology.org/2024.wmt-1.4/`.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL `https://arxiv.org/abs/2305.14314`.

Naman Goyal, Cynthia Gao, Vishrav Chaudhary, Peng-Jen Chen, Guillaume Wenzek, Da Ju, Sanjana Krishnan, Marc'Aurelio Ranzato, Francisco Guzmán, and Angela Fan. The Flores-101 evaluation benchmark for low-resource and multilingual machine translation. *Transactions of*

*the Association for Computational Linguistics*, 10:522–538, 2022. doi: 10.1162/tacl_a_00474. URL https://aclanthology.org/2022.tacl-1.30/.

Andrey Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. The unreasonable ineffectiveness of the deeper layers. *arXiv preprint arXiv:2403.17887*, 2024. URL https://arxiv.org/abs/2403.17887.

Daniel Han, Michael Han, and Unsloth team. Unsloth. https://github.com/unslothai/unsloth, 2023.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2106.09685.

Amir Hossein Kargaran, Ayyoob Imani, François Yvon, and Hinrich Schuetze. GlotLID: Language identification for low-resource languages. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6155–6218, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.410. URL https://aclanthology.org/2023.findings-emnlp.410/.

NLLB Team, Marta R. Costa-jussà, James Cross, Onur Çelebi, Maha Elbayad, et al. No language left behind: Scaling human-centered machine translation. *arXiv preprint arXiv:2207.04672*, 2022. URL https://arxiv.org/abs/2207.04672.

Maja Popović. chrF++: words helping character n-grams. In Ondřej Bojar, Christian Buck, Rajen Chatterjee, Christian Federmann, Yvette Graham, Barry Haddow, Matthias Huck, Antonio Jimeno Yepes, Philipp Koehn, and Julia Kreutzer, editors, *Proceedings of the Second Conference on Machine Translation*, pages 612–618, Copenhagen, Denmark, September 2017. Association for Computational Linguistics. doi: 10.18653/v1/W17-4770. URL https://aclanthology.org/W17-4770/.

Asahi Ushio, Yi Zhou, and Jose Camacho-Collados. Efficient multilingual language model compression through vocabulary trimming. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 14725–14739, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.981. URL https://aclanthology.org/2023.findings-emnlp.981/.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://arxiv.org/abs/2306.05685.